

A Practical Guide to Graph attention network

TOPIC -- GRAPH ATTENTION NETWORK

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

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Attention in Machine Learning is a technique that mimics cognitive attention. In the context of learning on graphs, the attention coefficient α_{uv} measures how important node v is to node u . Normalized attention coefficients are computed as follows:

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Graph neural networks (GNNs) are specialized artificial neural networks that are designed for tasks whose inputs are graphs. One prominent example is molecular drug design. Each input sample is a graph representation of a molecule, where atoms form the nodes and chemical bonds between atoms form the edges. In addition to the graph representation, the input also includes known chemical properties for each of the atoms. Dataset samples may thus differ in length, reflecting the varying numbers of atoms in molecules, and the varying number of bonds between them. The task is to predict the efficacy of a given molecule for a specific medical application, such as eliminating *E. coli* bacteria. The key design element of GNNs is the use of pairwise message passing, such that graph nodes iteratively update their representations by exchanging information with their neighbors. Several GNN architectures have been proposed, which implement different flavors of message passing, started by recursive or convolutional constructive approaches. As of 2022, it is an open question whether it is possible to define GNN architectures "going beyond" message passing, or instead every GNN can be built on message passing over suitably defined graphs.

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where $\mathbf{p}$ is a learnable projection vector. The projection vector $\mathbf{p}$ computes a scalar projection value for each graph node. The top-k pooling layer can then be formalised as follows:

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To represent an image as a graph structure, the image is first divided into multiple patches, each of which is treated as a node in the graph. Edges are then formed by connecting each node to its nearest neighbors based on spatial or feature similarity. This graph-based representation enables the application of graph learning models to visual tasks. The relational structure helps to enhance feature extraction and improve performance on image understanding.

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where $\mathbf{a}$ is a vector of learnable weights, $\cdot^T$ indicates transposition, $\mathbf{e}_{uv}$ are the edge features (if present), and LeakyReLU is a modified ReLU activation function. Attention coefficients are then normalized via softmax to make them easily comparable across different nodes. A GCN can be seen as a special case of a GAT where attention coefficients are not learnable, but fixed and equal to the edge weights w_{uv} .

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When viewed as a graph, a network of computers can be analyzed with GNNs for anomaly detection. Anomalies within provenance graphs often correlate to malicious activity within the network. GNNs have been used to identify these anomalies on individual nodes and within paths to detect malicious processes, or on the edge level to detect lateral movement.

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Graph-based representation of text helps to capture deeper semantic relationships between words. Many studies have used graph networks to enhance performance in various text processing tasks such as text classification, question answering, Neural Machine Translation (NMT), event extraction, fact verification, etc.

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where ϕ and ψ are differentiable functions (e.g., artificial neural networks), and $\bigoplus$ is a permutation invariant aggregation operator that can accept an arbitrary number of inputs (e.g., element-wise sum, mean, or max). In particular, ϕ and ψ are referred to as update and message functions, respectively. Intuitively, in an MPNN computational block, graph nodes update their representations by aggregating the messages received from their

neighbours. The outputs of one or more MPNN layers are node representations $\mathbf{h}_u$ for each node $u \in V$ in the graph. Node representations can be employed for any downstream task, such as node/graph classification or edge prediction. Graph nodes in an MPNN update their representation aggregating information from their immediate neighbours. As such, stacking n MPNN layers means that one node will be able to communicate with nodes that are at most n "hops" away. In principle, to ensure that every node receives information from every other node, one would need to stack a number of MPNN layers equal to the graph diameter. However, stacking many MPNN layers may cause issues such as oversmoothing and oversquashing. Oversmoothing refers to the issue of node representations becoming indistinguishable. Oversquashing refers to the bottleneck that is created by squeezing long-range dependencies into fixed-size representations. Countermeasures such as skip connections (as in residual neural networks), gated update rules and jumping knowledge can mitigate oversmoothing. Modifying the final layer to be a fully-adjacent layer, i.e., by considering the graph as a complete graph, can mitigate oversquashing in problems where long-range dependencies are required. Other "flavours" of MPNN have been developed in the literature, such as graph convolutional networks and graph attention networks, whose definitions can be expressed in terms of the MPNN formalism.

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$$\mathbf{h}_u = \phi \left(\mathbf{x}_u, \bigoplus_{v \in N_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{e}_{uv}) \right)$$

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where $i = \text{top}_k(y)$ is the subset of nodes with the top- k highest projection scores, $\odot$ denotes element-wise matrix multiplication. The self-attention pooling layer can be seen as an extension of the top- k pooling layer. Differently from top- k pooling, the self-attention scores computed in self-attention pooling account both for the graph features and the graph topology.

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$$\mathbf{H} = \sigma \left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{X} \Theta \right)$$

Message passing layers are permutation-equivariant layers mapping a graph into an updated representation of the same graph. Formally, they can be expressed as message passing neural networks (MPNNs). Let $G = (V, E)$ be a graph, where V is the node set and E is the edge set. Let N_u be the neighbourhood of some node $u \in V$. Additionally, let $\mathbf{x}_u$ be the features of node $u \in V$, and $\mathbf{e}_{uv}$ be the features of edge $(u, v) \in E$. An MPNN layer can be expressed as follows: